

Most of our late deliveries were never late

An end-to-end analysis of 180,519 order lines across 36 months finds that the delivery crisis is largely a promise-setting failure, while the real margin leak sits somewhere nobody was looking.

180,519
order lines analysed

Jan 2015 – Jan 2018
observation window

\$35.21M
sales in scope

164
destination countries

Executive report
prepared for operations and commercial leadership

On-time performance by service tier

Share of shipments that arrived within the window the customer was promised. Bar length shows on-time delivery; the figure at right is the miss rate.

57.29%

of all shipments recorded late

First Class

promised 1 day · delivered 2.00 days

100.00%

Second Class

promised 2 days · delivered 3.99 days

79.83%

Same Day

promised 0 days · delivered 0.48 days

47.93%

Standard Class

promised 4 days · delivered 3.99 days

39.77%

Standard Class is the only tier whose promise matches its actual performance, and it is also the only tier with a defensible miss rate. The three premium tiers all promise faster than the network has ever delivered — in **36 consecutive months**, across **all 23 regions**, without a single exception.

01 What the data says

Five findings change what leadership should do next. The first two reframe a problem the business has been treating as operational. The third relocates the profit conversation entirely.

The late-delivery rate is mostly an artefact of how delivery windows are defined

Every one of the 26,513 First Class shipments in the dataset was recorded late — a 100.00% miss rate in all 23 regions, in every month. That uniformity is not what operational failure looks like; operational failure varies. First Class is scheduled at 1 day and the network delivers it in 2.00 days, consistently and reliably. The shipments are not late. The promise is wrong.

Second Class repeats the pattern more loosely: promised in 2 days, delivered in 3.99, missing 79.83% of the time. Re-baselining these two tiers to the performance the network actually and dependably achieves would cut the headline late rate from 57.29% to roughly 34% without changing a single physical process.

Only one service tier has a genuine execution problem

Same Day promises 0 days, averages 0.48, and misses 47.93% of the time — but unlike the other tiers its performance swings hard by geography, from a 20.3% miss rate in Central Asia to 66.7% in Southern Africa. That variance is the signature of a real operational constraint, and it is the one delivery problem worth spending money to fix.

Loss-making orders destroy half of gross profit, and nobody is looking at them

139,354 order lines generated \$7.51M in profit. Another 32,295 lines — 18.69% of volume — lost \$3.71M. Those losses erase 49.35% of everything the profitable business earned, taking the company from a 21.34% margin to the 10.81% it actually books.

The asymmetry is the alarming part. The average losing line loses \$114.83; the average winning line earns \$53.93. Losses are 2.13 times more severe than wins. This is not thin-margin drift at the edges — it is a concentrated, structural leak that no category, region or segment report would ever surface, because every one of those aggregates shows a near-uniform 10–11% margin.

Late delivery is not what is costing the money

A delayed line earns \$21.62 on average; an on-time line earns \$22.58. The gap is 96 cents on a \$204 order — 4.3%. Across every delay bucket from two days early to four days late, mean profit stays inside a band of roughly \$20 to \$23. Whatever the late-delivery problem is costing, it is not showing up in order economics, and any business case built on recovering margin from faster shipping will not survive contact with this data.

That does not make delivery performance unimportant. It makes it a retention and reputation exposure rather than a margin one — and it means the fix should be cheap, because the payback cannot be measured in unit economics.

Revenue is concentrated to a degree that is itself a risk

Ten products out of 118 generate 90.0% of all sales. One department, Fan Shop, accounts for 46.6%. A single product — the Field & Stream Sportsman 16 Gun Fire Safe — carries 18.8% of company revenue on its own. The catalogue has 118 items; the business effectively has ten.

One caution before anything is forecast from this data

Monthly sales hold flat at roughly \$1.0M for 34 straight months, then fall to \$0.60M in November 2017, \$0.48M in December and \$0.32M in January 2018 — a 68% collapse in the final quarter with no corresponding change in order mix, delay rate, discounting or margin. A genuine commercial collapse of that speed would leave fingerprints elsewhere in the data. It leaves none. Treat the last three months as suspected incomplete capture and exclude them from any trend or forecast until the extract is verified at source.

02 Scope, data foundation and what was excluded

The analysis rests on a single transactional extract at order-line grain. Establishing what is in it — and what was deliberately taken out — determines how far the findings can be pushed.

The extract

The source file carries 180,519 rows across 53 columns, one row per order line, spanning 1 January 2015 to 31 January 2018. There are no duplicate rows and no duplicate order-line identifiers, so the grain is clean and no de-duplication was required. Those 180,519 lines roll up to 65,752 distinct orders placed by 20,652 customers, drawn from a catalogue of just 118 products in 50 categories and 11 departments, shipped to 164 countries across 23 regions and 5 markets.

180,519 Order lines	65,752 Distinct orders	20,652 Customers	118 Products in catalogue	50 Categories	164 Destination countries
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Average order size is 2.75 lines. Products carry 75 distinct price points and order quantities take only 5 distinct values, which tells us this is a fairly rigid catalogue with limited pricing flexibility — a constraint that matters later when we look at where margin is lost.

Columns removed before analysis

Nine columns were dropped, leaving 44. Two were empty or near-empty and two more were structurally unusable:

Columns removed, with the reason each was excluded

Column	MissingReason for removal
Product Description	100.00%Entirely empty — zero non-null values in 180,519 rows
Order Zipcode	86.24%Too sparse to support geographic analysis at postcode level
Customer Email	0%Masked to a single repeated placeholder — no information content
Customer Password	0%Masked to a single repeated placeholder; should not be in an analytical extract at all
Customer Fname, Lname	0%Personally identifying, analytically redundant against Customer Id
Customer Street, Zipcode	0%Personally identifying; city and state retained for geography
Product Image	0%Asset URL, not analytical data

The exclusion that shapes every delivery figure

7,754 records — 4.30% of the file — carry a delivery status of "Shipping canceled". These were removed before any delivery or profitability analysis, on the reasoning that a shipment that never shipped cannot be early, late or on time, and including it would contaminate both the denominator of the delay rate and the profit pool.

This is the right call, but it has a consequence worth stating plainly: **2,855 orders disappear from the analysis entirely**, and with them 391 customers. Cancellations are themselves a commercial signal — one that this analysis does not examine and that should be picked up separately. Nothing in this report describes why 4.3% of shipments were cancelled or what they were worth.

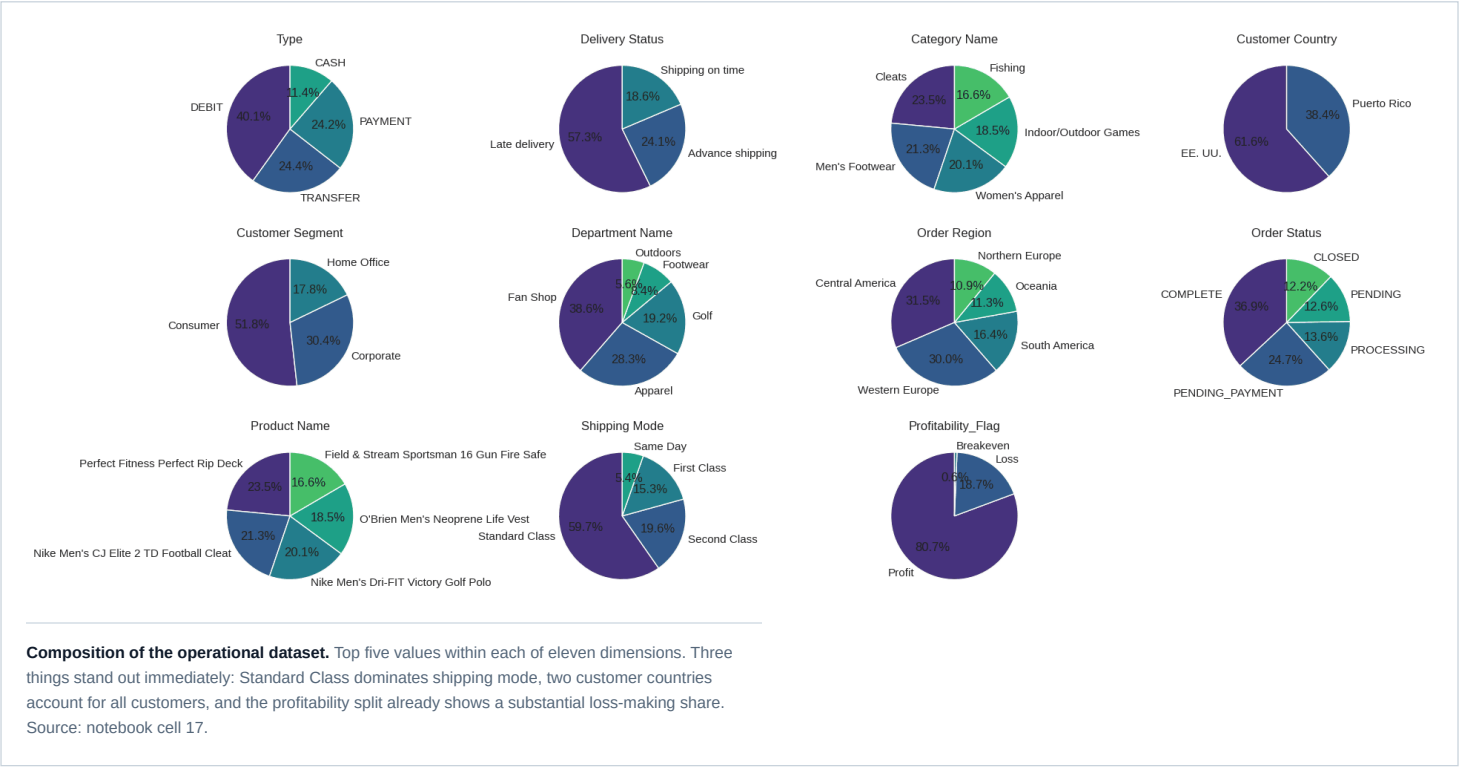
180,519 Rows received	-7,754 Shipping cancelled, removed	-0 Removed for missing essential fields	172,765 Operational rows analysed	62,897 Orders in scope	20,261 Customers in scope
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After the cancellation filter, no rows at all were lost to missing values in the nine fields required for analysis — sales, profit, delivery status, shipping mode, scheduled and actual transit days, and order date were complete across all 172,765 remaining records. For a file of this size that is unusually clean, and it means the findings below are not shaped by imputation choices.

Derived measures

Six fields were constructed. The central one is shipping delay, defined as actual transit days minus scheduled transit days, with any positive value flagged as a late delivery. A profitability flag splits each line into profit, loss or breakeven. Calendar attributes and a percentage margin complete the set.

One property of this construction deserves attention, because it validates the whole delivery analysis: the derived late-delivery flag produces exactly 98,977 records, which matches the source system's own "Late delivery" status count precisely. The independently computed measure and the system-of-record label agree on every single row. The delay figures in this report are therefore not an analytical interpretation — they are the operational reality as the business itself records it.



The customer base is geographically narrow in a way the destination data disguises. Customers sit in only 2 countries across 46 states and 563 cities, while orders ship to 164 countries. This is a concentrated seller shipping globally, not a global operation — which matters for how the delivery findings should be interpreted, since origin-side variation is effectively absent.

03 Commercial baseline

Before diagnosing anything, the headline position across the 36-month window.

\$35.21M Total sales	\$3.81M Net profit	\$3.71M Gross losses absorbed	10.81% Net profit margin	57.29% Late delivery rate	3.50 days Average actual transit
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2.93 days Average scheduled transit	1.62 days Average overrun when late	
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Two numbers on that board should be read together and rarely are. The network takes 3.50 days to deliver and promises 2.93. The gap is 0.57 days — the business systematically promises about half a day faster than it delivers, and has done so for three years. That single half-day is the mechanical source of a 57.29% late rate.

The second number is the overrun. When a shipment is late, it is late by 1.62 days on average. Very few shipments are catastrophically late. The problem is enormous in frequency and trivial in severity, which is precisely the profile of a calibration error rather than a capability failure.

How to read the rest of this report

Sections 4 through 6 examine delivery performance and conclude that most of it is a definition problem. Sections 7 through 9 examine where money is actually made and lost. Section 10 sets out what to do, ordered by return on effort rather than by size of the underlying number.

04

Delivery performance: anatomy of a 57% failure rate

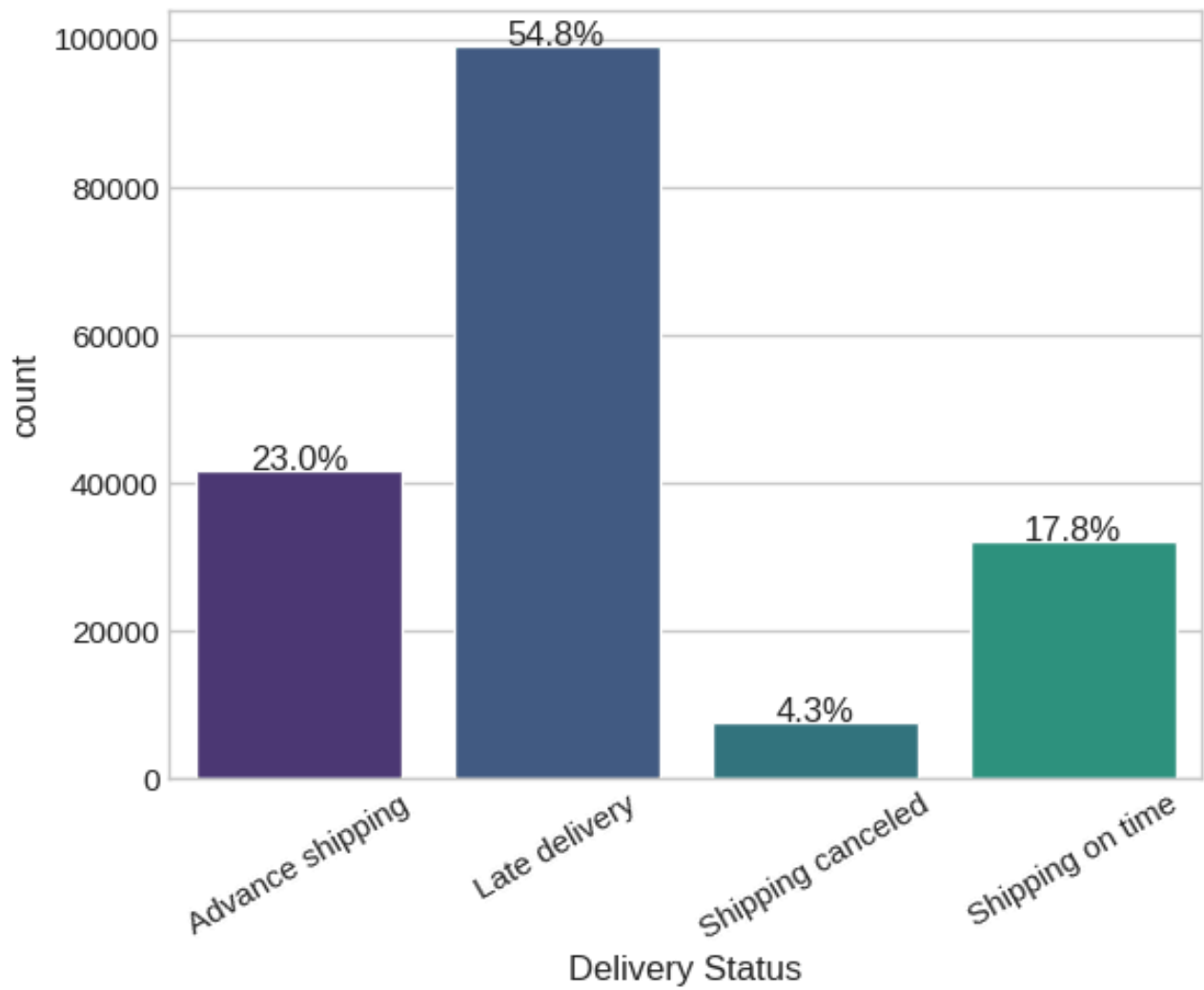
A miss rate that high normally means something is broken. Following it through four levels of detail shows that almost nothing is.

The shape of the problem

Delivery outcomes across 172,765 operational order lines

Outcome	Lines	Share	Orders	Sales	Profit	Avg delay
Late delivery	98,977	57.29%	36,048	\$20,126,395	\$2,140,052	+1.62 d
Advance shipping	41,592	24.07%	15,127	\$8,518,008	\$935,225	−1.50 d
Shipping on time	32,196	18.64%	11,722	\$6,570,027	\$731,144	0.00 d
Total	172,765	100%	62,897	\$35,214,430	\$3,806,421	+0.57 d

Note the second row. 24.07% of shipments arrive *early*, by an average of 1.50 days. A network that delivers early a quarter of the time and late 57% of the time is not an unreliable network — it is a network whose schedule has been set in the wrong place relative to a fairly stable transit-time distribution.



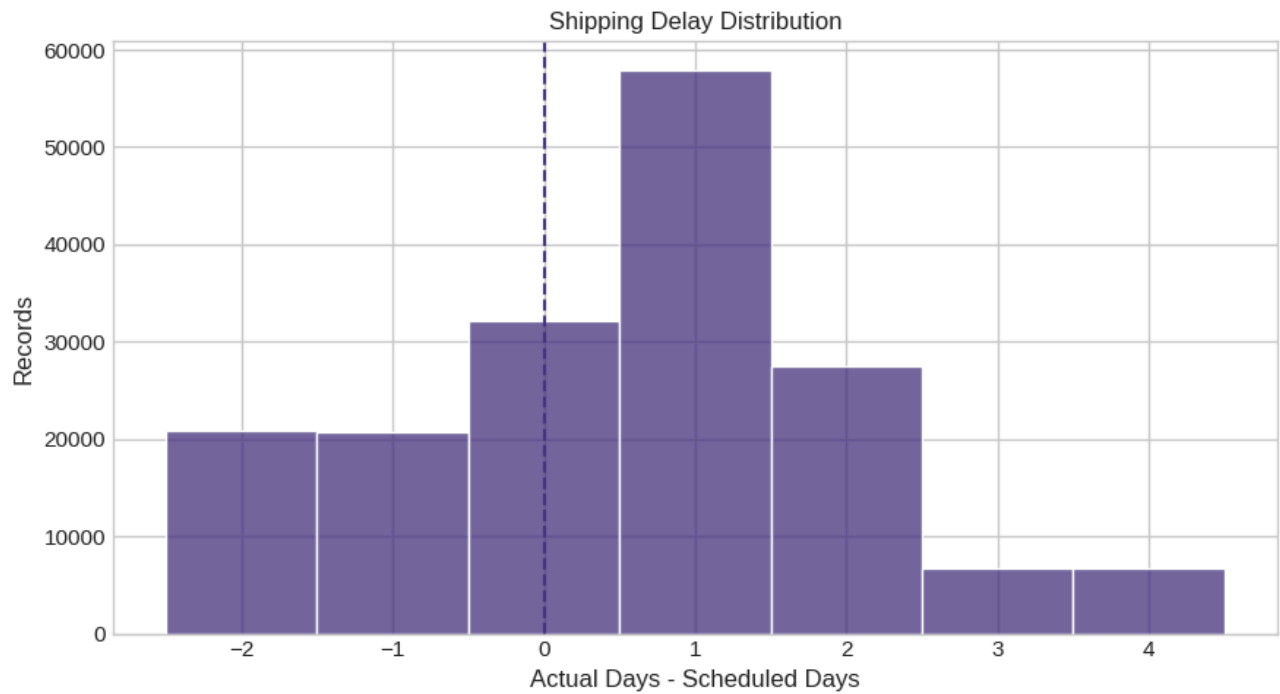
Delivery status distribution across the full dataset. Includes the 7,754 cancelled shipments excluded from all subsequent analysis. Source: notebook cell 20.

Delay severity

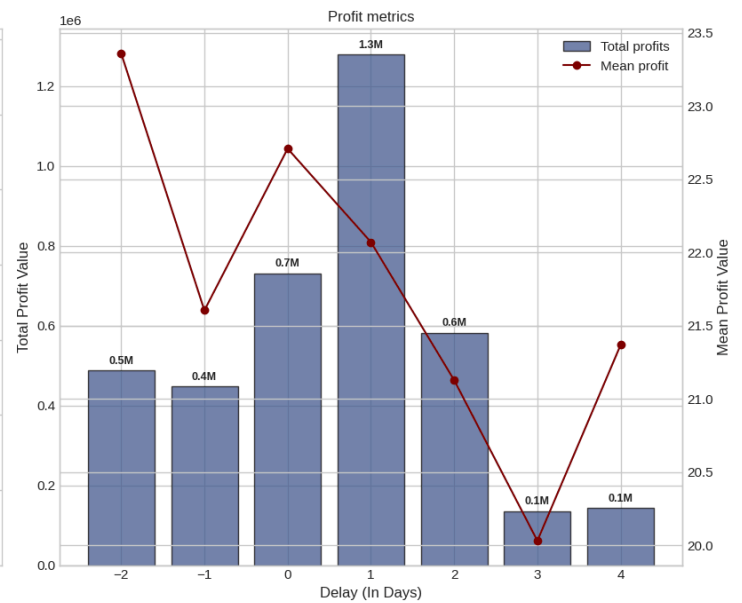
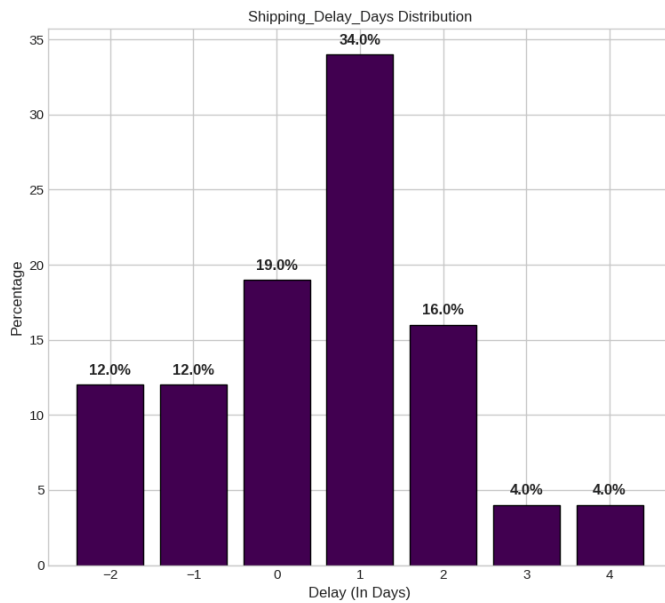
Distribution of shipping delay, in days, where negative values indicate early arrival

Delay	Lines	ShareReading
-2 days	20,873	12.1%Two days early
-1 day	20,719	12.0%One day early
0 days	32,196	18.6%Exactly as promised
+1 day	57,957	33.5%The single largest outcome in the dataset
+2 days	27,551	15.9%Together with +1 day, 49.5% of all volume
+3 days	6,772	3.9%Genuine service failure
+4 days	6,697	3.9%Genuine service failure
Total	172,765	100%

The distribution is tight and bounded. Nothing arrives more than 4 days late or more than 2 days early; the entire operation lives inside a 7-day span. Delays of 3 or 4 days — the ones a customer would actually escalate — account for 13,469 lines, or 7.8% of volume. The other 85,508 late shipments, 86.4% of all late events, are late by one or two days against a promise the network was never positioned to keep.



Shipping delay distribution. The dashed line marks the promised date. The mass of the distribution sits one day to its right. Source: notebook cell 23.



Delay distribution against profit. The left panel restates delay frequency. The right panel is the one that matters: total profit (bars) simply tracks volume, while mean profit (line) stays inside a narrow band of roughly \$20 to \$23 across every delay level, including four days late. **Delay severity has no meaningful relationship with order profitability.** Source: notebook cell 24.

The finding that should stop a project

If late delivery destroyed value at the order level, mean profit would fall as delay rises. It does not. Comparing all delayed lines against all on-time lines: \$21.62 versus \$22.58 in average profit, on average sales of \$203.34 versus

\$204.48, at identical quantities (2.13 units) and identical discount rates (10%).

Any initiative justified on the basis of recovering margin through faster delivery has no support in this dataset. The justification for fixing delivery has to be customer retention and brand exposure — which are real, but are not measured here and should not be conflated with margin recovery.

05 **Where the delay actually comes from**

Slicing the delay rate by every available dimension produces one clear answer and a great deal of noise. The noise is instructive.

Service tier is the only dimension that discriminates

Delivery performance by shipping mode, ordered by miss rate

Mode	Lines	Scheduled	Actual	Gap	Miss rate	Sales	Profit
First Class	26,513	1.00 d	2.00 d	+1.00	100.00%	\$5,408,069	\$616,597
Second Class	33,806	2.00 d	3.99 d	+1.99	79.83%	\$6,860,662	\$722,861
Same Day	9,293	0.00 d	0.48 d	+0.48	47.93%	\$1,854,872	\$194,182
Standard Class	103,153	4.00 d	3.99 d	-0.01	39.77%	\$21,090,827	\$2,272,780

Read the gap column. Standard Class is calibrated to within a hundredth of a day and misses 39.77% of the time — that residual is the network's genuine variability, the irreducible noise of moving physical goods. Every other tier's miss rate is that same variability plus a systematic offset baked into the promise.

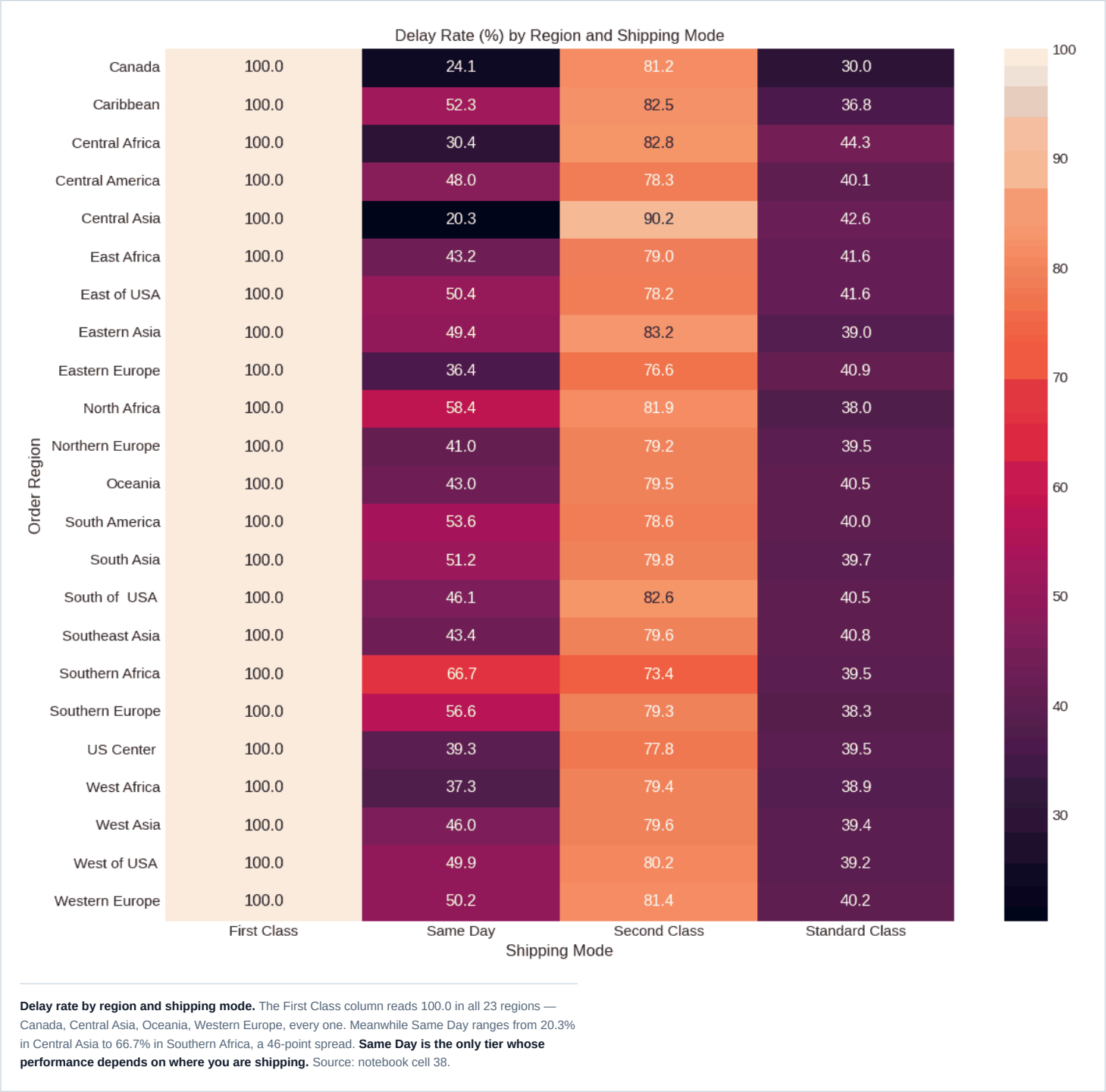
First Class is the extreme case, and the arithmetic is worth stating explicitly. Its mean actual transit is exactly 2.00 days and 100% of shipments exceed the 1-day schedule, which means every First Class shipment took precisely 2 days. Not approximately — precisely. The tier is the most predictable service in the entire network, and it is recorded as a total failure because it is measured against a promise it has never once met in three years and 26,513 shipments.



Late delivery rate by shipping mode. The ranking is inverted from intuition: the faster the promised service, the worse the recorded performance. That is the signature of promises outrunning capability, not of premium logistics failing. Source: notebook cell 31.

The confirmation: it holds everywhere, without exception

If the First Class problem were operational, it would vary by geography. Some regions would manage it; others would not. The region-by-mode matrix settles the question.



Two different problems, wearing the same label

First Class and Second Class fail uniformly across every region and month. Uniform failure against a fixed threshold is a definition problem, and it is fixed by changing the definition.

Same Day fails variably — 20.3% in one region, 66.7% in another, against the same 0-day promise and the same network. Variable failure is an execution problem, and it is fixed by changing the operation. Its 46-point regional spread is where genuine logistics investigation belongs.

Everything else is flat

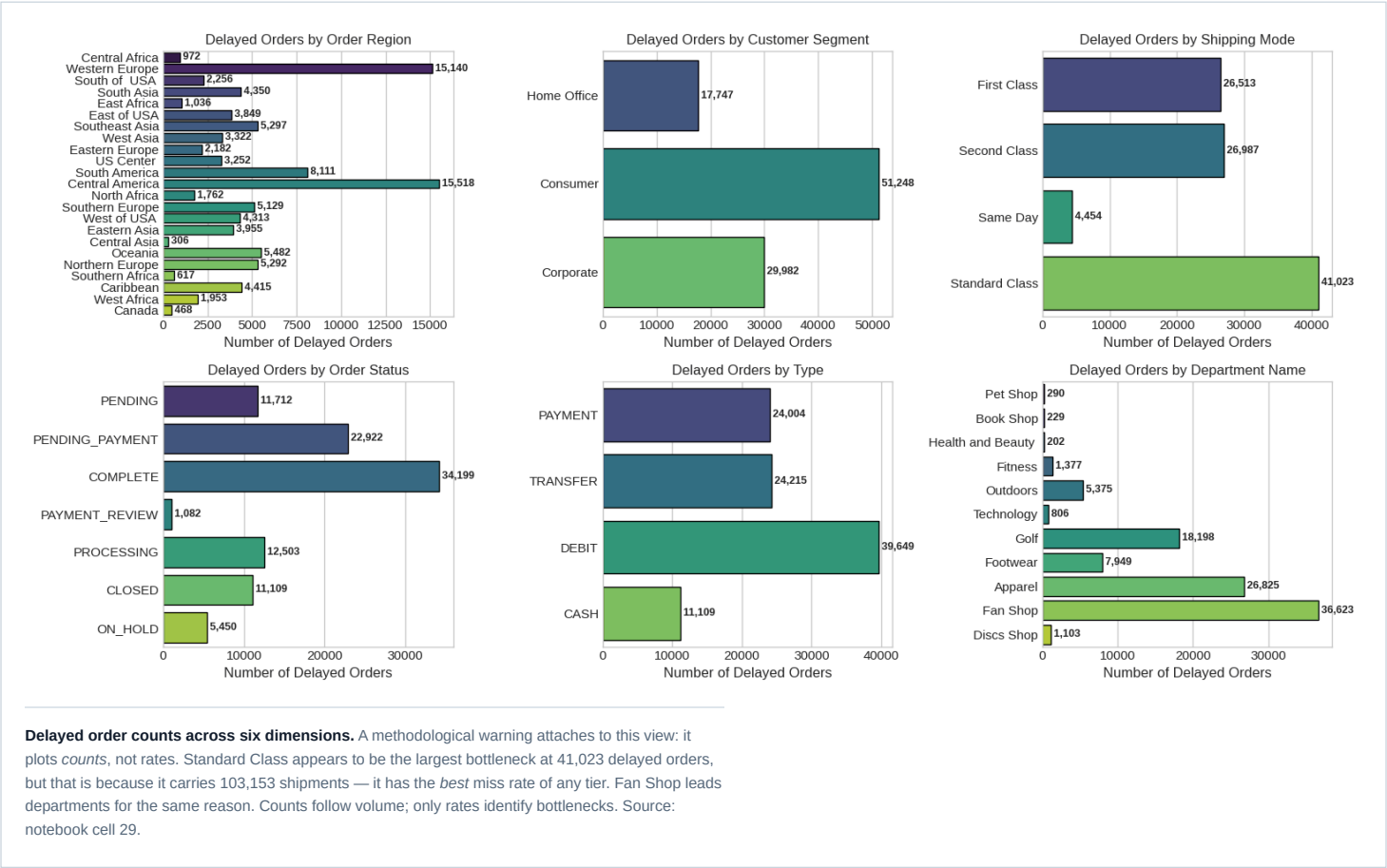
The remaining dimensions were tested and none discriminates meaningfully:

Spread in late-delivery rate across each analytical dimension

Dimension	Levels	Lowest	Highest	SpreadVerdict
Shipping mode	4	39.77%	100.00%	60.2 ptsThe entire signal
Order region	23	56.09%	60.15%	4.1 ptsEffectively uniform
Department	11	56.77%	61.44%	4.7 ptsEffectively uniform
Customer segment	3	57.08%	57.59%	0.5 ptsIndistinguishable
Category	50	~56%	68.85%	~13 ptsTop values are small-sample artefacts

The category spread deserves a caveat rather than a strategy. The worst-performing category, Golf Bags & Carts at 68.85%, rests on 61 records. Lacrosse at 62.61% rests on 329. Once sample sizes reach the thousands, every category converges on 56–59%. These are not problem categories; they are small numbers behaving like small numbers.

The same applies to regions. Central Africa tops the regional table at 60.15% on 1,616 records, while Western Europe sits at 58.53% on 25,867. A 4-point spread across 23 regions carrying wildly different volumes is consistent with random variation around a single underlying rate, not with regional capability differences.

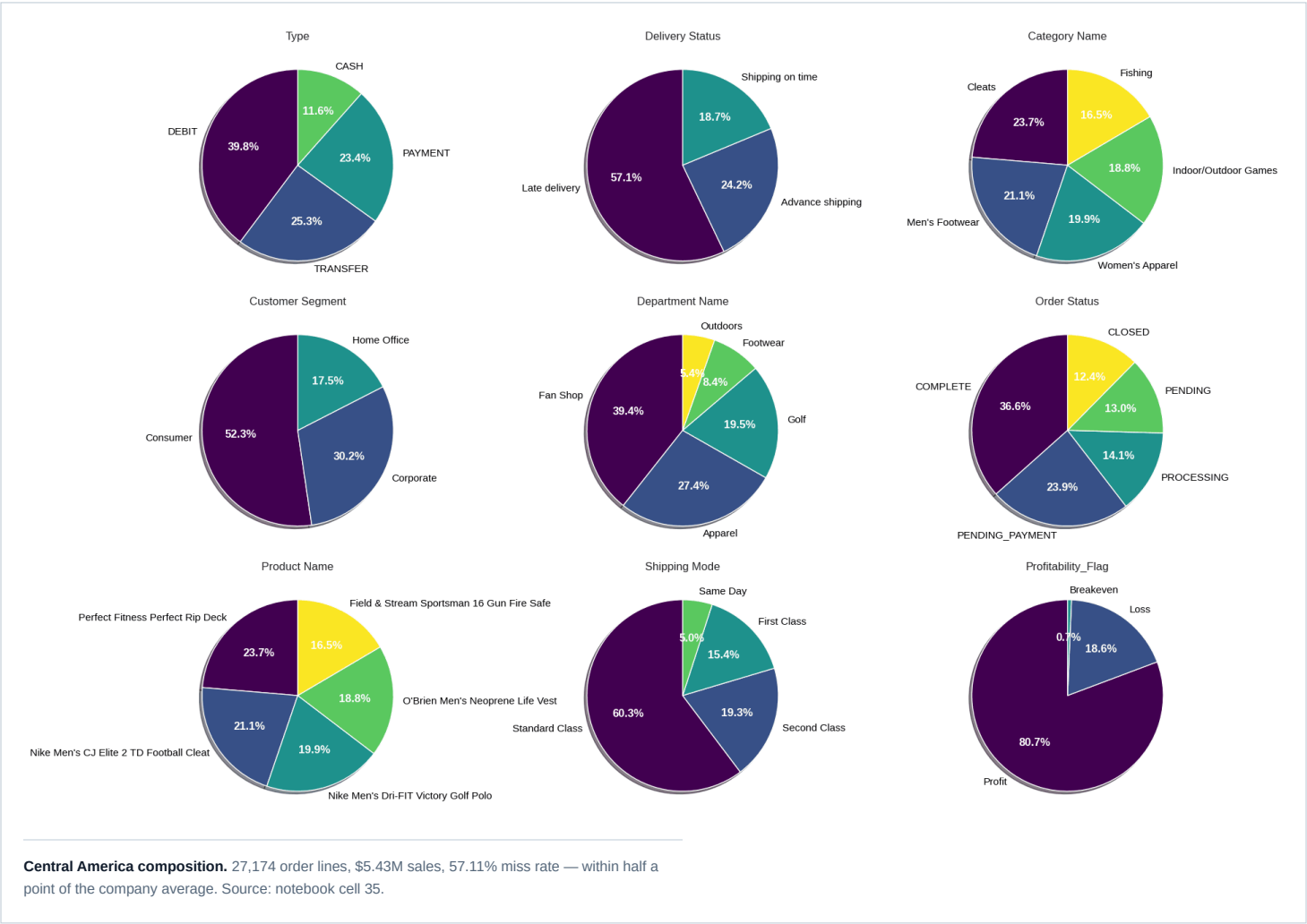


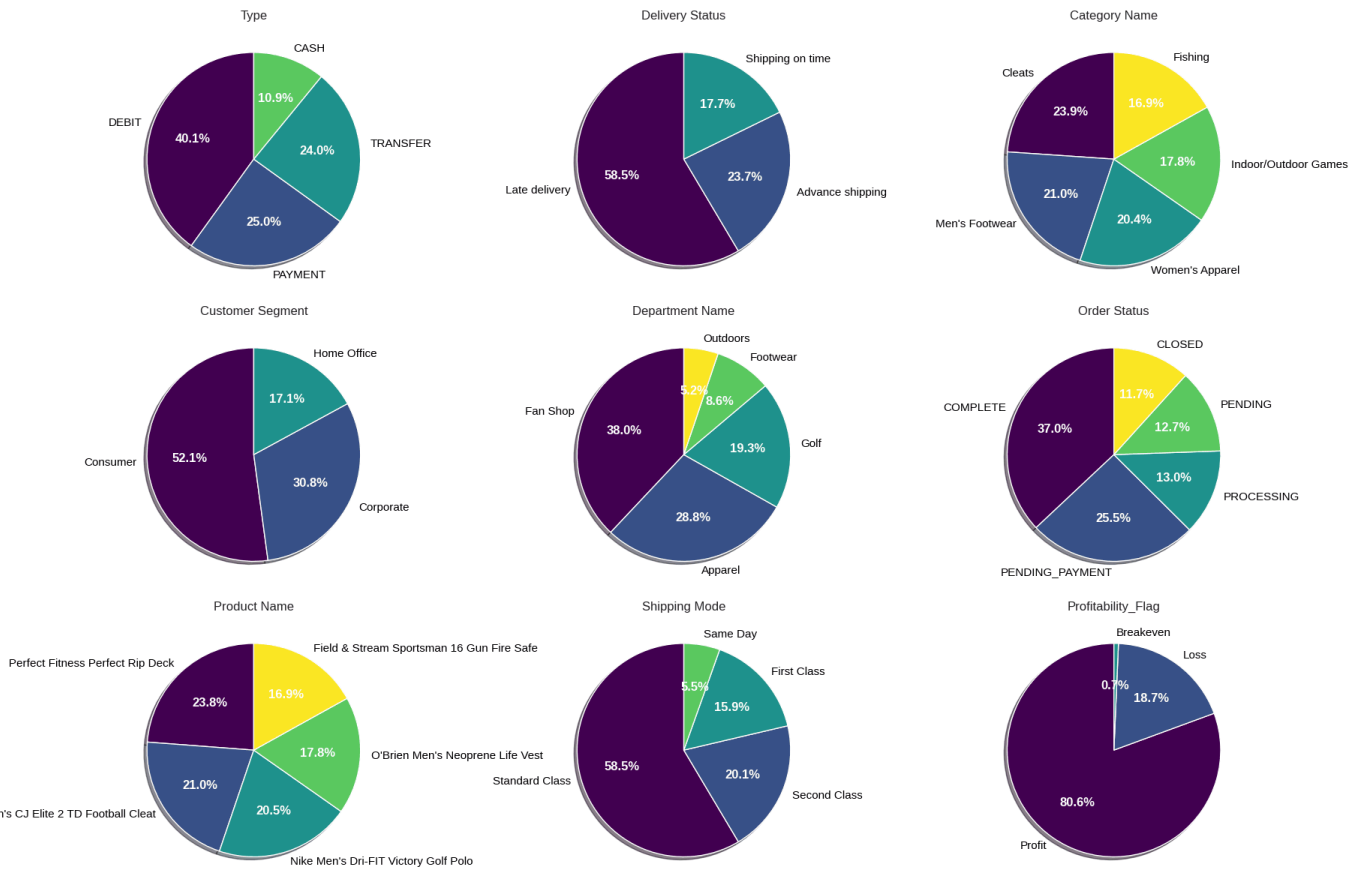
An analytical caution worth carrying forward

Volume-weighted count charts will reliably nominate the biggest part of the business as the biggest problem. In this dataset that produces exactly the wrong answer: the chart above points at Standard Class and Fan Shop, while the rate analysis points at First Class and Same Day. Any dashboard built from these figures should present rates as the default and counts only as a secondary measure of exposure.

Regional detail

The two largest regions by volume were profiled in full. Central America (27,174 lines, \$5.43M) and Western Europe (25,867 lines, \$5.63M) between them carry 30.7% of all order lines. Their composition is close to identical to each other and to the company as a whole — the same mode mix, the same segment mix, the same profitability split — which is further confirmation that regional variation is not where the story is.



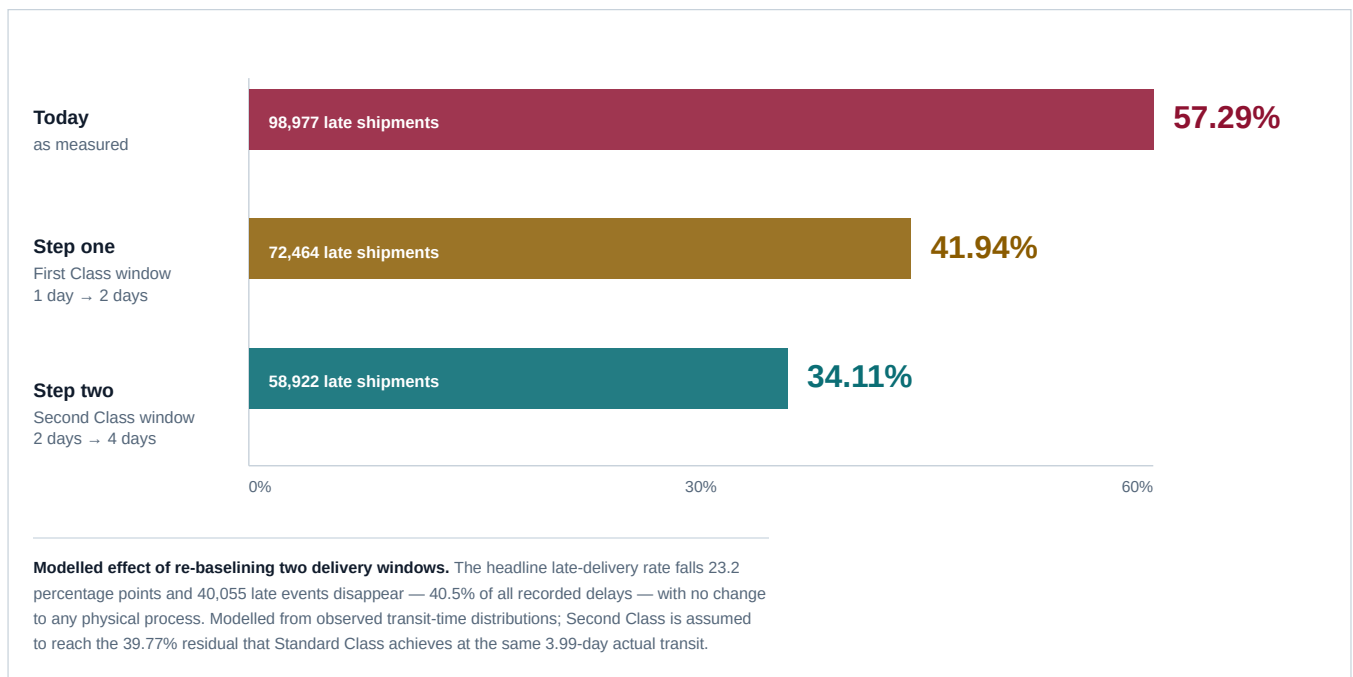


Western Europe composition. 25,867 order lines, \$5.63M sales, 58.53% miss rate. The largest region by revenue and, by a narrow margin, the worst of the two — but the difference between them is 1.4 percentage points, which is not a difference worth acting on. Source: notebook cell 36.

06 What re-baselining the promise would actually do

Two schedule changes, no operational spend, no new capability. The modelled effect on the headline metric is larger than any physical intervention available.

The logic is narrow and testable. First Class delivers in exactly 2.00 days, every time, everywhere; setting its window to 2 days takes its miss rate to zero. Second Class delivers in 3.99 days on average; setting its window to 4 days should bring it to the same 39.77% residual that Standard Class exhibits at an identical 3.99-day actual transit. Standard Class and Same Day are left untouched.



What this is, and what it is not

This is not a measurement trick to make a number look better. Customers currently receive a 1-day promise and a 2-day delivery, and the business has known for three years that it will be a 2-day delivery. Continuing to promise 1 day is the problem; it manufactures 26,513 broken promises a period and buries the 13,469 genuinely bad shipments inside them.

The commercial question re-baselining raises is real and should be faced: if First Class and Standard Class both take about 2 to 4 days, the premium tiers may not be differentiated enough to justify their price. That is a pricing and product question, and this data raises it clearly — First Class at 2.00 days and Second Class at 3.99 days against Standard Class at 3.99 days means **Second Class and Standard Class are the same service**. One of them should probably not exist.

After re-baselining, the residual 34.11% is the honest operating position, and it is made of two things: the network's genuine day-to-day variability, and the Same Day failure rate. Only the second is addressable, and it affects 9,293 lines carrying \$1.85M in sales — 5.3% of revenue.

07 Where the money is actually lost

The delivery investigation consumes most of the analysis and turns out to be largely a labelling exercise. The profitability investigation takes three lines and finds half the profit pool missing.



The profit bridge. Loss-making order lines absorb 49.35% of everything the profitable business earns. Breakeven lines — 1,116 of them on \$221,223 of sales — are omitted as they contribute exactly zero.

Order lines grouped by profitability outcome

Outcome	Lines	Share	Sales	Profit	Per line	Margin
Profit	139,354	80.66%	\$28,430,011	\$7,514,814	\$53.93	26.43%
Loss	32,295	18.69%	\$6,563,196	-\$3,708,393	-\$114.83	-56.50%
Breakeven	1,116	0.65%	\$221,223	\$0.00	\$0.00	0.00%
Total	172,765	100%	\$35,214,430	\$3,806,421	\$22.03	10.81%

The asymmetry is the whole problem

A losing line loses \$114.83. A winning line earns \$53.93. Losses are **2.13 times more severe** than wins, and they arrive on sales of only \$6.56M — meaning the business is destroying 56.50 cents of value on every dollar of revenue passing through that 18.69% of volume.

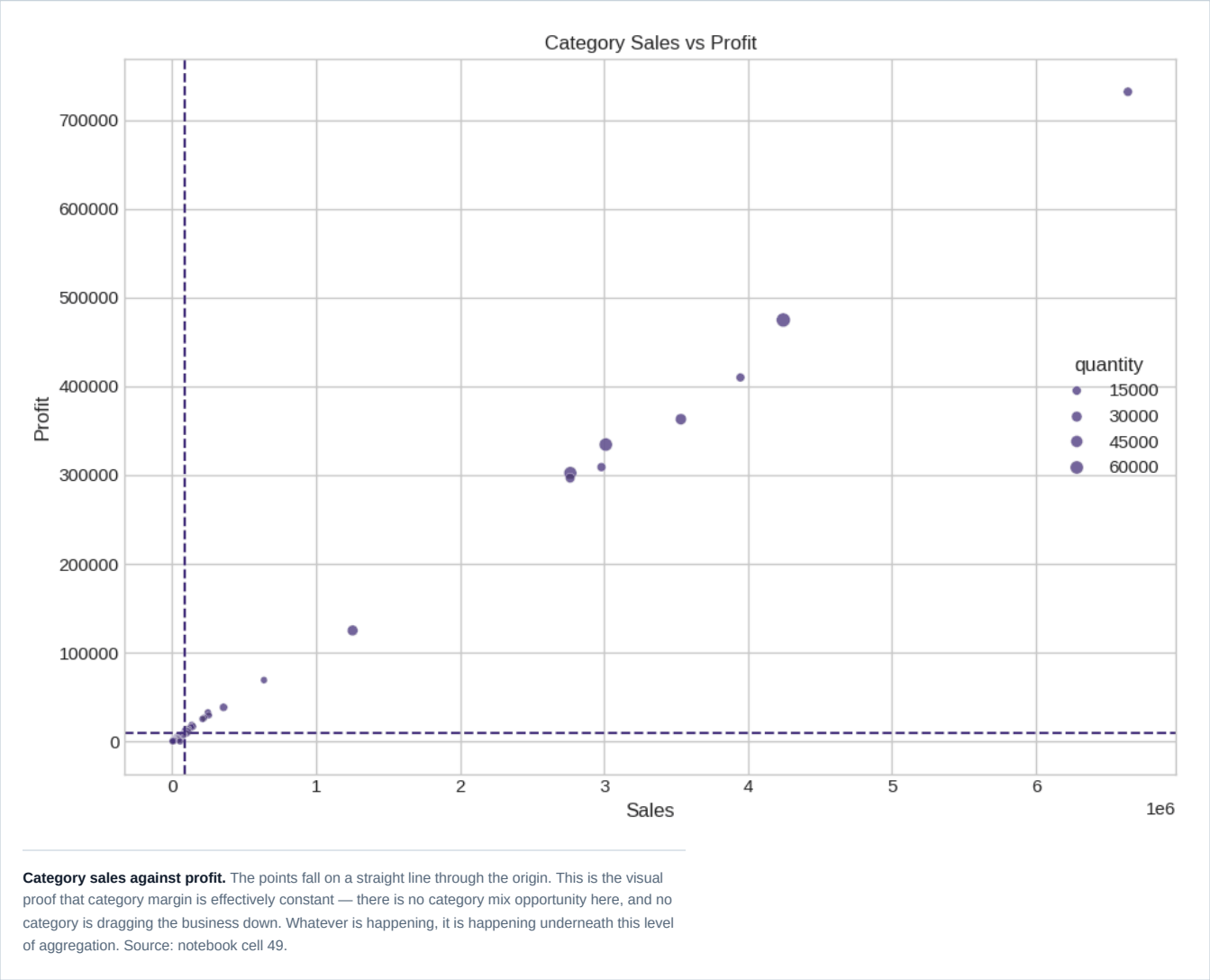
This is not a pricing rounding error. A loss of that magnitude relative to line value implies either products being sold well below landed cost, discounting stacked past the point of recovery, or a cost allocation the commercial team cannot see at the point of sale. Identifying which of those three it is, is the single highest-value question in this report — and it cannot be answered from the fields available here.

Why no aggregate report would ever find this

The loss concentration is invisible in every summary view the business is likely to run. Margins are remarkably uniform once aggregated:

- All three customer segments book margins of 10.67%, 10.81% and 10.86% — a range of 0.19 points.
- Every one of the top ten categories by revenue books between 9.98% and 11.18%.
- Every region books between roughly 10% and 11%.

The variance lives entirely at order-line level, where a \$114.83 loss sits next to a \$53.93 gain and the two average out to a respectable-looking 10.81%. Any investigation conducted at category, region or segment grain will conclude, correctly and uselessly, that everything is fine.



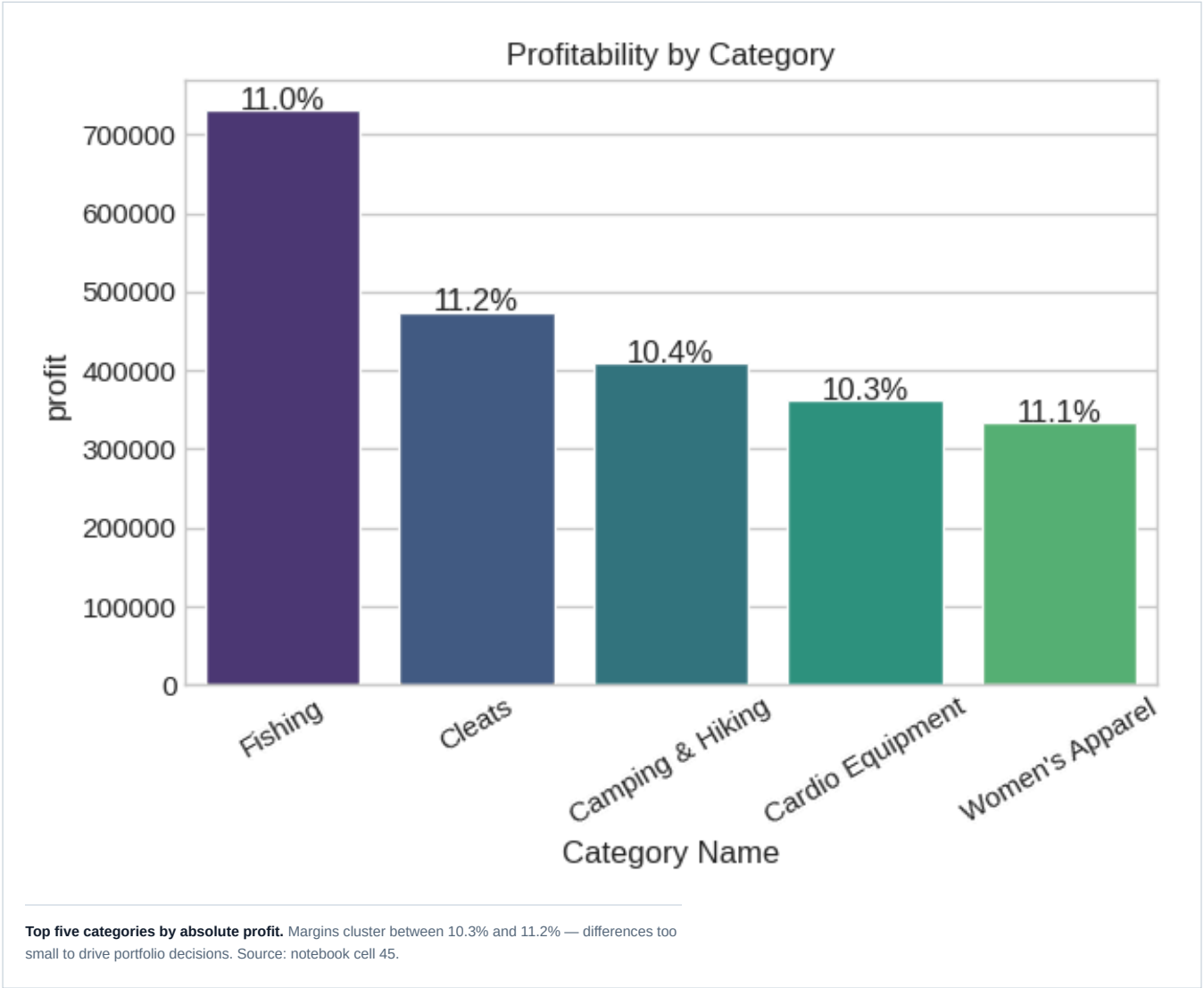
The marginal category exceptions

A median-split quadrant analysis places 24 categories in high sales / high profit, 24 in low sales / low profit, and isolates exactly two outliers. Given that near-perfect linearity, the quadrant framework has little to work with — but the low-margin tail is still worth a commercial review:

Weakest ten categories by profit margin, with the quadrant outlier highlighted

Category	Sales	Profit	Margin	Units
Strength Training	\$54,518	\$244	0.45%	109
Men's Clothing	\$41,748	\$1,829	4.38%	198
Books	\$12,152	\$854	7.03%	391
As Seen on TV!	\$19,898	\$1,449	7.28%	199
Basketball	\$26,199	\$1,910	7.29%	64
Video Games	\$31,959	\$2,521	7.89%	804
Pet Supplies	\$39,837	\$3,505	8.80%	472
Health and Beauty	\$101,392	\$9,281	9.15%	346
DVDs	\$75,450	\$7,169	9.50%	459
Boxing & MMA	\$81,727	\$7,928	9.70%	1,210

Strength Training is the clearest anomaly: \$54,518 of sales producing \$244 of profit, a 0.45% margin against a 10.81% company average. Health and Beauty is the sole occupant of the high sales / low profit quadrant at 9.15%. Both are small in absolute terms — together they are 0.44% of revenue — so they are governance flags rather than value opportunities. They do, however, demonstrate that the pricing or costing model can produce near-zero margin outcomes, which is consistent with what the loss-making lines suggest at scale.



Discounting: a real but second-order lever

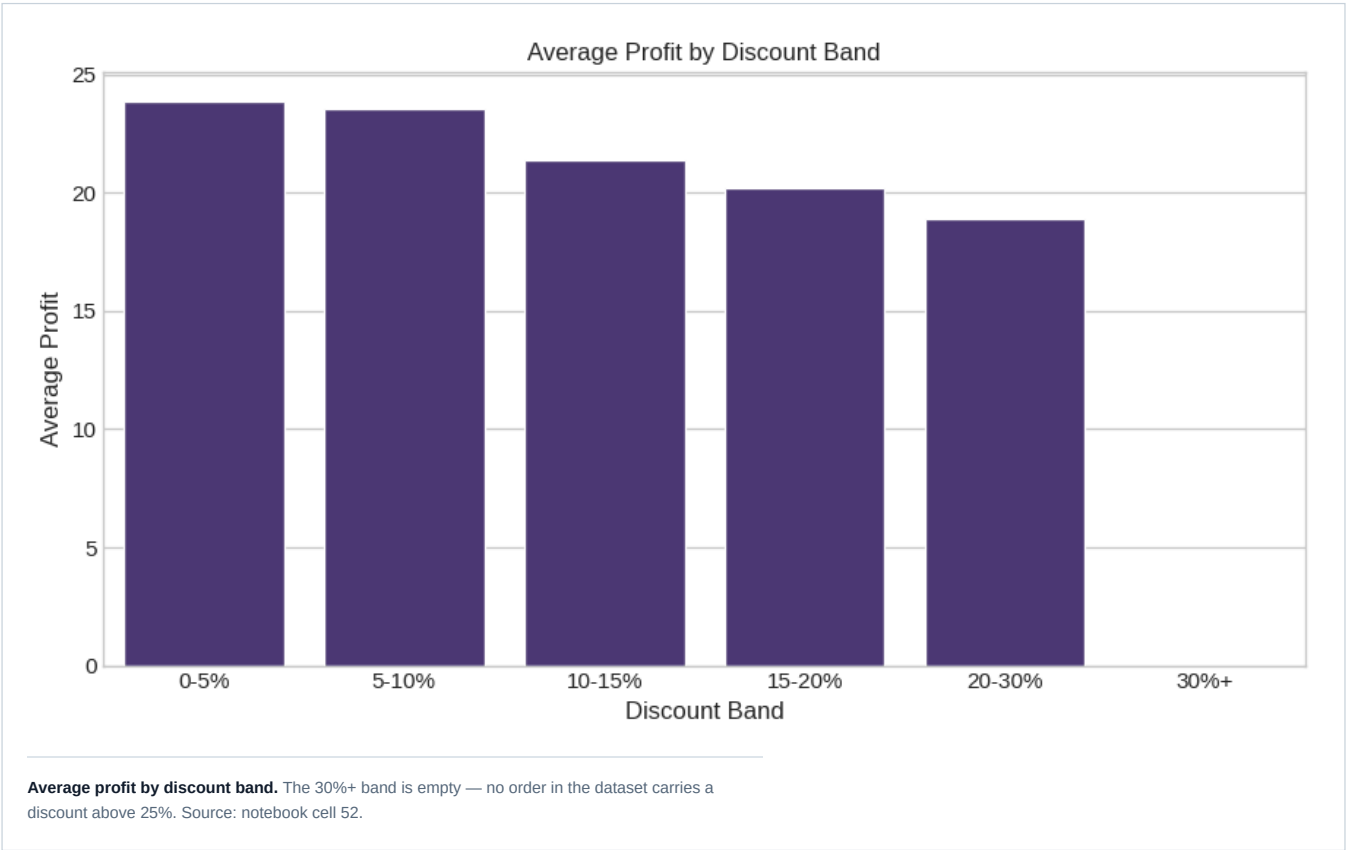
Average profit per line declines monotonically as discount depth increases, from \$23.89 in the shallowest band to \$18.88 in the deepest — a 21.0% erosion. The relationship is clean and well-behaved, with no inflection point, which suggests discounting is being applied as policy rather than opportunistically.

Profitability by discount band

Discount band	Lines	Share	Sales	Avg profit	Total profit
0–5%	48,024	27.8%	\$9,781,543	\$23.89	\$1,147,311
5–10%	38,418	22.2%	\$7,833,270	\$23.58	\$905,854
10–15%	28,717	16.6%	\$5,852,937	\$21.40	\$614,558
15–20%	38,404	22.2%	\$7,830,273	\$20.21	\$776,188
20–30%	19,202	11.1%	\$3,916,407	\$18.88	\$362,509
Total	172,765	100%	\$35,214,430	\$22.03	\$3,806,421

A third of all volume — 57,606 lines — carries a discount above 15%. The observed maximum discount is 25%, so the deepest band is not being abused; there is a ceiling and it is being respected. If the two deepest bands were brought to the profit-per-line of the 5–10% band, the recovery would be approximately \$220,000, or 5.8% of net profit.

That is worth having, but it is 6% of the value sitting in loss-making orders. Discount discipline should be a standing commercial practice, not the centrepiece of a margin programme.



08 Revenue concentration

A catalogue of 118 products conceals a business that depends on ten of them.

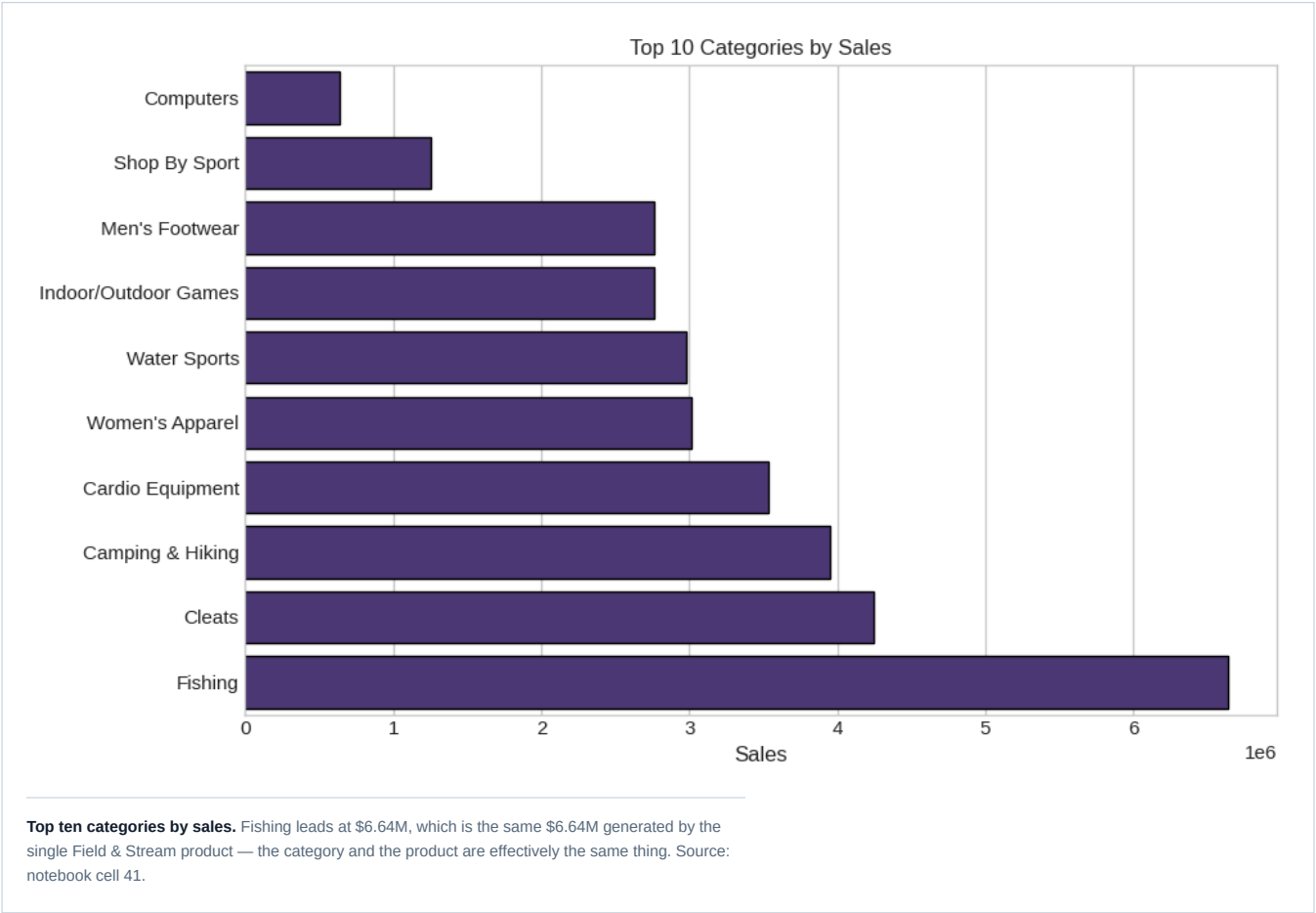
90.0% of sales from 10 of 118 products	46.6% of sales from one department	18.8% of sales from one product	60.7% of sales from 5 of 50 categories
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Top ten products by sales, which together account for 90.0% of company revenue

Product	Sales	Share	Profit	Units	Miss rate
Field & Stream Sportsman 16 Gun Fire Safe	\$6,637,668	18.85%	\$731,576	16,595	65.51%
Perfect Fitness Perfect Rip Deck	\$4,233,794	12.02%	\$473,821	70,575	69.07%
Diamondback Women's Serene Classic Comfort Bike	\$3,946,837	11.21%	\$409,896	13,157	63.52%
Nike Men's Free 5.0+ Running Shoe	\$3,507,549	9.96%	\$359,794	35,079	62.50%
Nike Men's Dri-FIT Victory Golf Polo	\$3,011,600	8.55%	\$334,336	60,232	67.16%
Pelican Sunstream 100 Kayak	\$2,967,852	8.43%	\$307,911	14,840	64.61%
O'Brien Men's Neoprene Life Vest	\$2,765,543	7.85%	\$302,404	55,333	66.40%
Nike Men's CJ Elite 2 TD Football Cleat	\$2,763,977	7.85%	\$296,711	21,263	67.51%
Under Armour Girls' Toddler Spine Surge Running Shoe	\$1,214,896	3.45%	\$121,834	30,380	62.39%
Dell Laptop	\$637,500	1.81%	\$69,232	425	52.71%
Ten products	\$31,687,218	89.98%	\$3,407,515	317,879	—

The miss-rate column carries a note of its own. These ten products show delay rates between 52.71% and 69.07%, above the 57.29% company average — but the metric here is computed per unique order rather than per line, so it is not directly comparable to the rates elsewhere in this report. Read the ranking, not the level.

Dell Laptop is the interesting entry. It sells 425 units for \$637,500 — \$1,500 per unit against a company average order line of \$204 — and carries the lowest miss rate of the top ten at 52.71%. It is the only high-value, low-volume product in the leading group, and it behaves differently from everything around it.



Department structure

Sales and delivery performance by department

Department	Lines	Sales	Share	Profit	Margin	Miss rate
Fan Shop	64,033	\$16,392,418	46.55%	\$1,759,601	10.73%	57.19%
Apparel	46,884	\$7,631,674	21.67%	\$845,530	11.08%	57.22%
Golf	31,768	\$4,407,982	12.52%	\$476,329	10.81%	57.28%
Footwear	13,891	\$3,833,832	10.89%	\$388,979	10.15%	57.22%
Outdoors	9,267	\$1,200,938	3.41%	\$141,337	11.77%	58.00%
Technology	1,396	\$994,974	2.83%	\$111,589	11.21%	57.74%
Fitness	2,374	\$380,160	1.08%	\$45,270	11.91%	58.00%
Discs Shop	1,943	\$219,070	0.62%	\$24,145	11.02%	56.77%
Health and Beauty	346	\$101,392	0.29%	\$9,281	9.15%	58.38%
Pet Shop	472	\$39,837	0.11%	\$3,505	8.80%	61.44%
Book Shop	391	\$12,152	0.03%	\$854	7.03%	58.57%

Four departments carry 91.6% of revenue. The remaining seven contribute 8.4% between them and include all three of the sub-9% margin departments. Health and Beauty, Pet Shop and Book Shop together generate \$153,381 in sales — 0.44% of the business — at margins materially below every other department. They are candidates for rationalisation on grounds of complexity cost rather than lost profit.

Geography

The commercial footprint is broad; the risk is not. The United States leads at \$4.66M, but no single country exceeds 13.2% of revenue and the top ten countries together account for 63.4%. Geographic diversification is genuinely good — it is the product portfolio, not the market portfolio, that is concentrated.

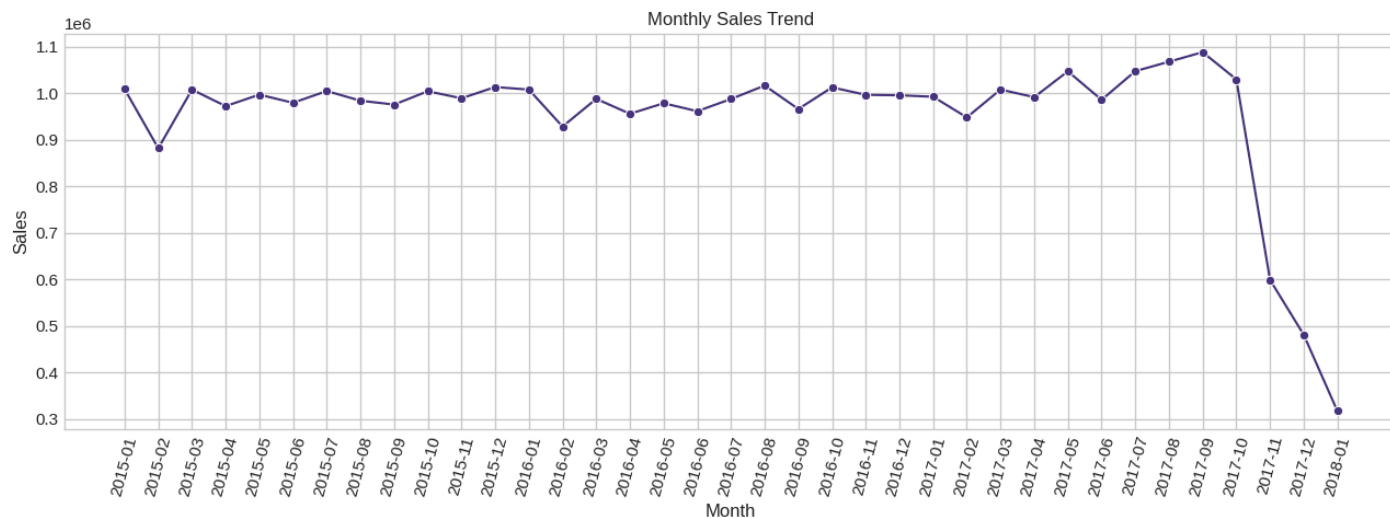
Top ten destination countries by sales. Country names appear as recorded in the source, in Spanish.

Country	Market	Orders	Sales	Share	Profit	Miss rate
Estados Unidos	USCA	7,887	\$4,659,618	13.23%	\$516,687	57.63%
Francia	Europe	4,648	\$2,753,971	7.82%	\$316,543	58.21%
México	LATAM	4,201	\$2,522,998	7.16%	\$289,018	57.49%
Alemania	Europe	3,368	\$1,986,194	5.64%	\$187,981	58.80%
Australia	Pacific Asia	3,633	\$1,623,924	4.61%	\$165,277	56.70%
Reino Unido	Europe	2,661	\$1,544,404	4.39%	\$174,689	55.82%
Brasil	LATAM	2,520	\$1,513,085	4.30%	\$175,033	57.14%
China	Pacific Asia	2,495	\$1,123,516	3.19%	\$117,357	56.79%
Italia	Europe	1,793	\$1,021,387	2.90%	\$116,446	56.10%
India	Pacific Asia	2,070	\$925,156	2.63%	\$94,874	58.49%

Miss rates across the top ten countries span 55.82% to 58.80% — a 3-point range around the company mean, again showing no geographic signal.

09 Trend over 36 months

Two time series, both of which say something the business should act on — one about data, one about the nature of the delivery problem.



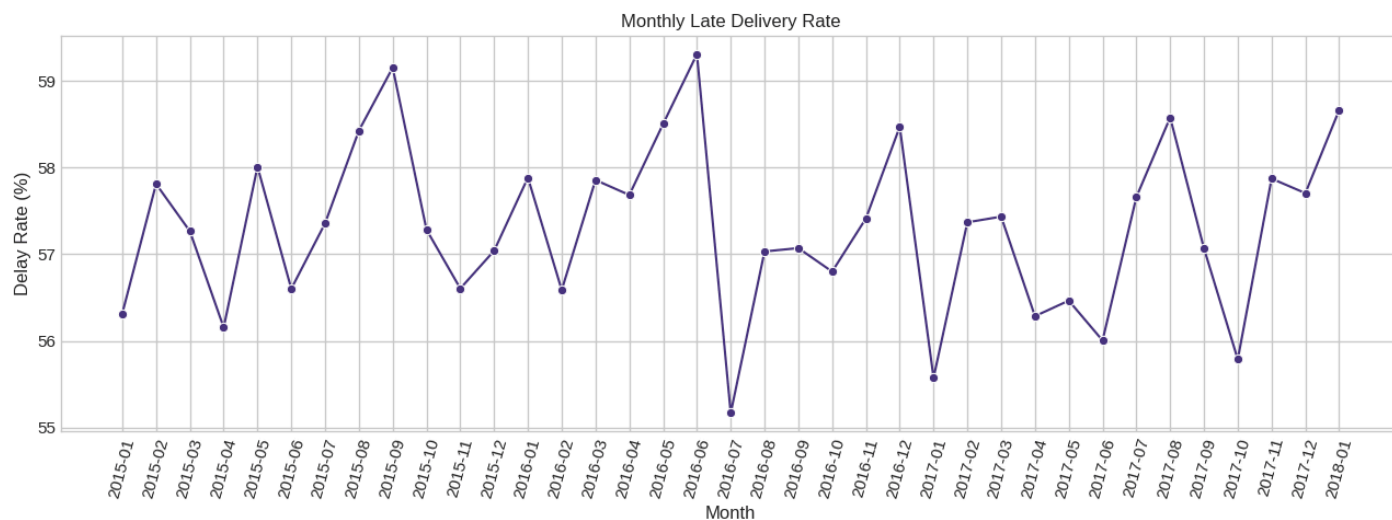
Monthly sales, January 2015 to January 2018. Thirty-four months of remarkable stability between roughly \$0.88M and \$1.09M, followed by a fall to \$0.60M, \$0.48M and \$0.32M. Source: notebook cell 62.

The final quarter is almost certainly incomplete, not collapsing

A real 68% revenue collapse over three months would show up in the composition of the data. It does not. Across the same period the late-delivery rate stays inside its normal band, in fact *rising* to roughly 58.7% in January 2018. Margin holds. Discount mix holds. Order size holds. The only thing that changes is the count of records.

That pattern — volume falling while every ratio stays constant — is the classic signature of a truncated extract or a reporting lag, not of a business losing customers. Until the extract is verified at source, November 2017 through January 2018 should be excluded from every trend line, forecast and year-on-year comparison. Reporting a 68% decline that turns out to be an ETL artefact is an expensive mistake to make in public.

Set the tail aside and the underlying series is notable for how little happens in it. Thirty-four months with no growth trend, no visible seasonality, and monthly values holding within about five percent of their mean. The opening months of the series run between 1,505 and 1,713 orders, and the band barely moves for three years. This is a business in a steady state, which is itself worth flagging: three years of flat revenue is a strategic fact, not a neutral backdrop.



Monthly late-delivery rate. Thirty-six months bounded between 55.18% and 59.31%, oscillating with no trend, no seasonality and no response to anything. Source: notebook cell 63.

The strongest single piece of evidence in this report

The late-delivery rate never once drops below 55% in 36 consecutive months, and never once exceeds 60%. It does not improve after any intervention, degrade in any peak, or respond to the collapse in volume at the end of the series. A metric that stable is not measuring performance — it is measuring a fixed relationship between two fixed numbers.

Those two numbers are a mean transit time of 3.50 days and a mean promise of 2.93. Until one of them moves, this line will not.

10 Customer segments

Included for completeness, and because the absence of a finding is a finding.

Performance across the three customer segments

Segment	Orders	Lines	Sales	Share	Profit	Margin	Avg line	Miss rate
Consumer	32,614	89,420	\$18,264,367	51.87%	\$1,982,864	10.86%	\$204.25	57.31%
Corporate	19,017	52,528	\$10,711,379	30.42%	\$1,157,821	10.81%	\$203.92	57.08%
Home Office	11,266	30,817	\$6,238,684	17.72%	\$665,736	10.67%	\$202.44	57.59%

The three segments differ in size and in nothing else. Margins span 0.19 percentage points. Average line value spans \$1.81. Miss rates span 0.51 points. Order composition, discount behaviour and category mix are effectively identical.

This has a practical implication. Segment-based differentiation — pricing, service levels, targeted retention — has no empirical foundation in this data, because the segments do not behave differently. Either the segmentation is not capturing real behavioural difference, or the business is treating three genuinely different customer types identically enough that they have converged. Both possibilities are worth investigating, and neither supports building a programme on this segmentation as it stands.

11 What to do

Ordered by return on effort rather than by the size of the number each one addresses. The first two cost nothing and move the most visible metric in the business; the third is where the money is.

1 Reset the First Class delivery window from 1 day to 2 days

The network delivers First Class in exactly 2.00 days, in every region, in every month, across 26,513 shipments and three years. It has never delivered one in a day. The current window guarantees a recorded failure on every single shipment and produces the 100.00% miss rate that dominates the headline number.

This is a change to a field in the order management system. It requires no capability, no capital and no supplier negotiation. It does require the commercial decision described in recommendation 2.

Effect: removes 26,513 late events, 26.8% of all recorded delays. Headline late rate falls from 57.29% to 41.94%. **Cost: none.**

2 Reset Second Class to 4 days, and decide whether it should exist

Second Class takes 3.99 days. Standard Class takes 3.99 days. They are the same service sold at two prices with two different promises, and the only consequence of the difference is that one of them is recorded as failing 79.83% of the time while the other fails 39.77%.

Re-baselining Second Class to 4 days brings it in line with Standard Class performance. But the underlying question is a product question: a tier structure where First Class is 2 days, Second Class is 4 days and Standard Class is also 4 days has one redundant tier. Consolidating would simplify the promise, the pricing and the operation at once.

Effect: removes a further ~13,465 late events. Combined with recommendation 1, the headline rate reaches 34.11% — a 23.2 point improvement and 40.5% of all delays eliminated. **Cost: none operationally; requires a pricing review.**

3 Open an investigation into the 32,295 loss-making order lines

This is the largest recoverable value in the business and it is not currently visible in any aggregate report, because every category, region and segment averages a healthy-looking 10–11% margin while a fifth of all volume destroys \$114.83 per line.

The dataset cannot say why. It can say that these losses are severe rather than marginal, that they are not explained by discounting (which erodes profit by only \$5.01 per line across its entire range), and that they are not explained by delivery performance (which moves profit by \$0.96). The cause is therefore in cost, price or product mix at a level of detail this extract does not carry. The first step is a sample review of the 500 worst lines against landed cost.

Effect: eliminating half the loss value adds **\$1.85M to net profit, a 48.7% increase**. Eliminating a quarter adds \$927,098, a 24.4% increase. **Cost: analyst time plus access to cost data.**

4 Fix Same Day — the only genuine execution failure

Same Day promises 0 days and averages 0.48, missing 47.93% of the time. Unlike every other tier, its performance varies sharply by geography: 20.3% in Central Asia against 66.7% in Southern Africa, a 46-point spread on the same promise and the same network. That variance means the problem is operational and therefore fixable.

Start with the worst regions. If Southern Africa and North Africa were brought to the performance Central Asia and Canada already achieve, the tier's miss rate would roughly halve.

Scope: 9,293 lines, \$1.85M sales, 5.3% of revenue. **Effect:** up to ~2,200 late events removed. **Cost:** genuine operational investment — the only recommendation here that requires it.

5 Verify the November 2017 to January 2018 extract before anything is forecast

Revenue falls 68% across the final three months while every ratio in the business holds constant. That pattern is far more consistent with incomplete data capture than with commercial collapse, but it cannot be confirmed from the extract alone.

Until it is confirmed at source, exclude the tail from all trend analysis. If the decline turns out to be real, it is the most urgent item in this report by a wide margin and everything else should be re-sequenced behind it.

Effect: prevents a reporting error with significant external consequences, or surfaces an emergency. **Cost:** one conversation with the data owner.

6 Tighten discount governance above 15%

A third of volume carries discounts above 15%, and profit per line falls steadily with depth. There is no evidence of abuse — the observed ceiling is 25% and is respected — so this is a matter of tightening a working policy rather than repairing a broken one.

Effect: approximately \$220,000, or 5.8% of net profit, if the two deepest bands reach 5–10% band performance. **Cost:** policy change.

7 Treat product concentration as a board-level risk

Ten of 118 products generate 90.0% of revenue, one department generates 46.6%, and one product generates 18.8%. Nothing in this data is going wrong as a result — but the loss of any single top-five product would remove 8 to 19% of company revenue, and no mitigation currently exists.

Effect: risk mitigation, not profit. **Cost:** strategic attention.

8 Analyse the 7,754 cancelled shipments this study excluded

Cancellations were correctly removed from the delivery analysis, but they were then not examined at all. They represent 4.30% of all records, 2,855 orders and 391 customers who appear nowhere in this report. Cancellation is a commercial signal — it may correlate with the same conditions that produce loss-making orders, and it is currently unmeasured.

Effect: closes a blind spot covering 4.3% of activity. **Cost:** a follow-on analysis.

12 Data quality register and methodology

Every finding above depends on the extract behaving as expected. Here is where it does not, and what was done about it.

Data quality issues identified, with their effect on the analysis

Issue	Detail	Effect on findings
Final quarter volume collapse	Sales fall 68% across Nov 2017 – Jan 2018 while every ratio holds constant	Material. All trend analysis and forecasting is unsafe until verified
Chart exports are blank files	All 15 saved PNG files are empty 640×480 white images	Material to reporting. See the note below
Cancellations unexamined	7,754 records (4.30%), 2,855 orders, 391 customers removed and never analysed	Blind spot covering 4.3% of activity
Breakeven lines are exactly zero	1,116 lines produce \$221,223 of sales and precisely \$0.00 profit	Exact zeros at this scale suggest a default or placeholder rather than a genuine outcome
Product Description empty	100% missing across all 180,519 rows	None — column dropped
Order Zipcode sparse	86.24% missing	Prevents postcode-level geographic analysis
Credentials in analytical extract	Customer Email and Password present, masked to a single placeholder value	None analytically; a governance concern that these fields are in the extract at all
Zero-variance column	Product Status holds one value across every row	None — carries no information
Mixed-language geography	Country names recorded in Spanish; customer country appears as "EE.UU."	Will break joins against standard country reference data
Redundant delay fields	Late_delivery_risk and Delivery Status encode the same information as the derived delay flag	Beneficial — the three agree exactly, which validates the delay measurement
Minor nulls	Customer Lname (8 rows), Customer Zipcode (3 rows)	None — fields dropped

Every chart the notebook saved to disk is blank

All fifteen exported PNG files are empty white images of 640×480 pixels, containing a single colour and no content. The cause is consistent across all fifteen: `plt.show()` is called before `plt.savefig()`. In a notebook backend, `show()` renders and then clears the current figure, so `savefig()` writes the blank canvas that remains, at default size rather than the figure size specified.

The charts reproduced in this report were recovered from the base64 image data embedded in the notebook's own cell outputs, so they are the correct figures at full resolution. But anyone who opened the exported files — in a deck, a shared folder or a report — received blank images. Reversing the two calls in all fifteen cells fixes it.

Methodological notes

- **Grain.** All analysis is at order-line level unless stated. Order-level figures are reported separately where they differ materially, and the two should not be mixed — 172,765 lines represent 62,897 orders, a ratio of 2.75.
- **The delay definition.** Late means actual transit days exceed scheduled transit days by any amount. A shipment one day late and a shipment four days late

count identically in the headline rate, which is why severity is reported separately in section 4.

- **Counts against rates.** Several charts in the source notebook plot counts of delayed orders by dimension. Counts track volume and will always nominate the largest part of the business. Rate analysis is used throughout this report and reaches the opposite conclusion on which tier and which department are problematic.
- **Small samples.** Category-level delay rates reaching 68.85% rest on as few as 61 records. No conclusion in this report rests on a category or region cell below 1,000 records.
- **Product miss rates.** The product table computes delay rate against unique orders rather than order lines, which inflates the level relative to every other rate in this report. The ranking is valid; the absolute values are not comparable.
- **The re-baselining model.** First Class reaching a 0% miss rate follows directly from the observed data: mean actual transit of exactly 2.00 days with 100% of shipments exceeding a 1-day schedule means every shipment took exactly 2 days. Second Class reaching 39.77% is an assumption — it is the rate Standard Class achieves at the same 3.99-day actual transit, and it should be validated before the change is committed.
- **What this data cannot answer.** Why orders lose money; what cancellations cost; whether late delivery affects repeat purchase; whether the final quarter is real. The first and the last are the two questions worth pursuing first.

Scope. 180,519 order lines from the DataCo Global supply chain extract, covering 1 January 2015 to 31 January 2018. 172,765 lines analysed after removing 7,754 cancelled shipments. All monetary figures are as recorded in the source and are not inflation-adjusted.

Sources. Analysis reproduced from the supply chain analytics notebook; figure captions cite the originating cell. Charts were recovered from embedded notebook outputs after the exported files were found to be blank. Modelled figures in section 6 and recommendations 1 through 3 are identified as estimates where they rest on assumption rather than observation.